

PORTFOLIO PROJECT

DATA ANALYTICS

Instacart Customer Analytics Dashboard

End-to-End Data Analytics Project using Python, DuckDB, SQL & Power BI
Transforming 3.4M+ transactions into executive-level business insights

Author: Abdul Samhoon | **Date:** 2026

Business Problem & Objectives

The Challenge

Online grocery platforms generate millions of daily transactions, making it difficult to extract meaningful, actionable insights from raw data. Businesses need clarity on:

- Customer purchasing behavior
- Product demand patterns
- Inventory planning
- Customer retention

Project Objectives

Analytical Database

Build a fast analytical layer using DuckDB

Purchasing Behavior

Analyze ordering patterns and trends

Product Performance

Evaluate departments, aisles, and demand

Executive Dashboards

Deliver interactive Power BI dashboards

Dataset & Technology Stack

3.4M+
Orders

134
Aisles

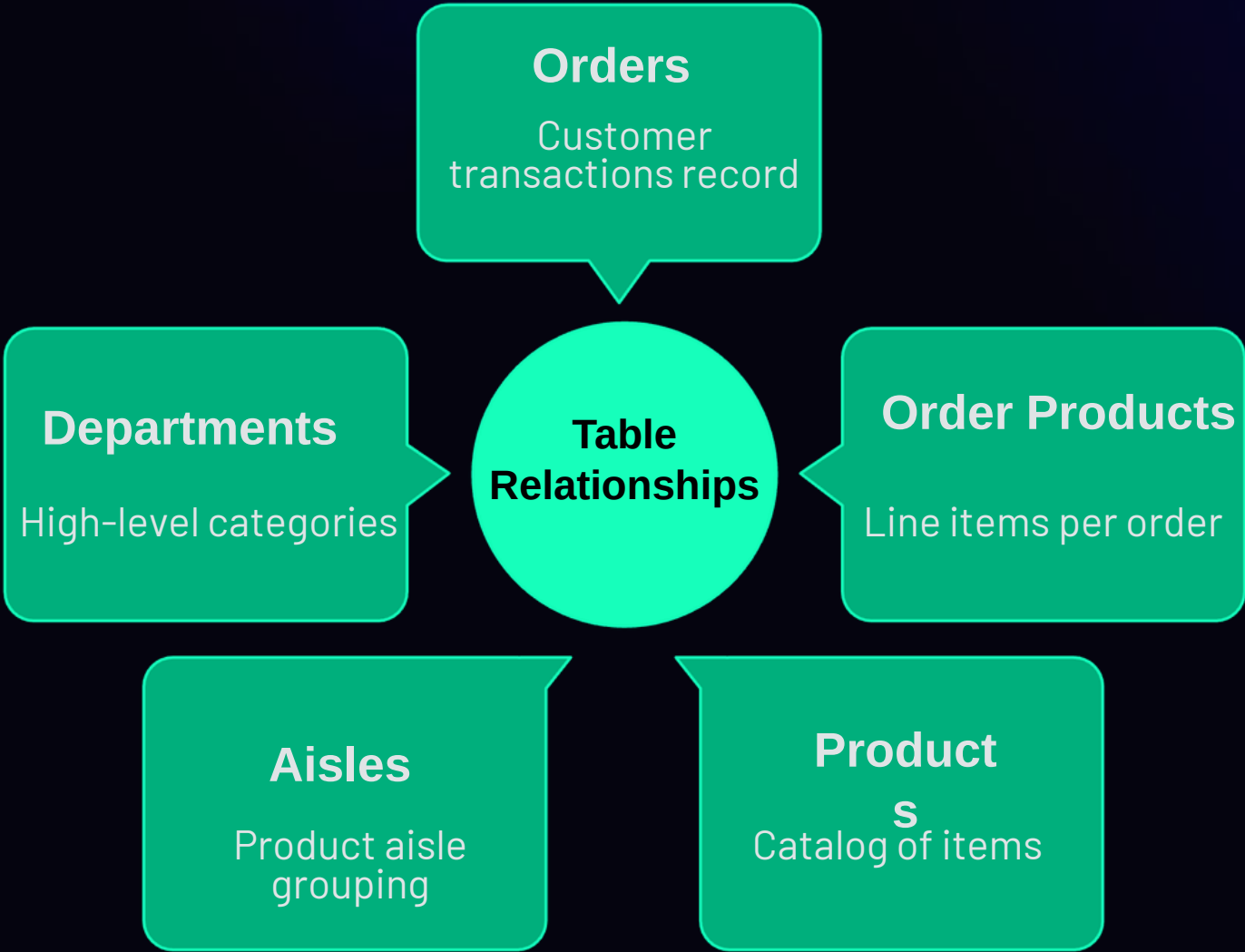
206K
Customers

21
Departments

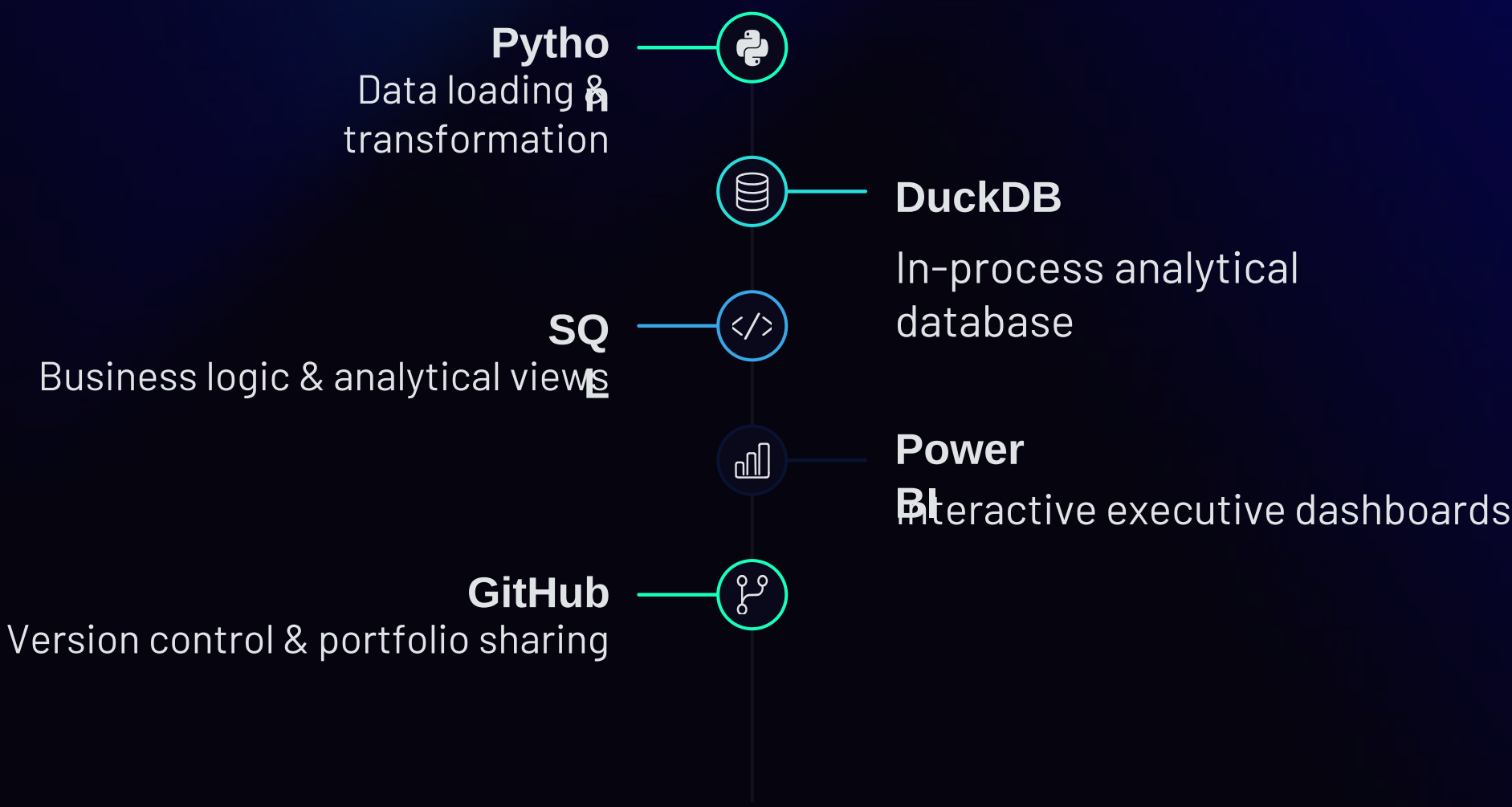
49K
Products

Dataset Schema

Five interconnected tables form the relational backbone of the analysis.



Technology Stack



End-to-End Analytics Pipeline

From raw CSV files to executive-ready dashboards — a reproducible workflow built for scale and clarity.



Ingest CSV

Load with Python

Store in DuckDB

SQL Analysis

Business Questions Solved Using SQL

1

Peak Ordering Hours

When do customers order most?
Identified daily demand curves.

2

Top Purchased Products

Which products drive volume?
Ranked by order frequency.

3

Department Performance

Which departments perform best?
Measured by revenue share.

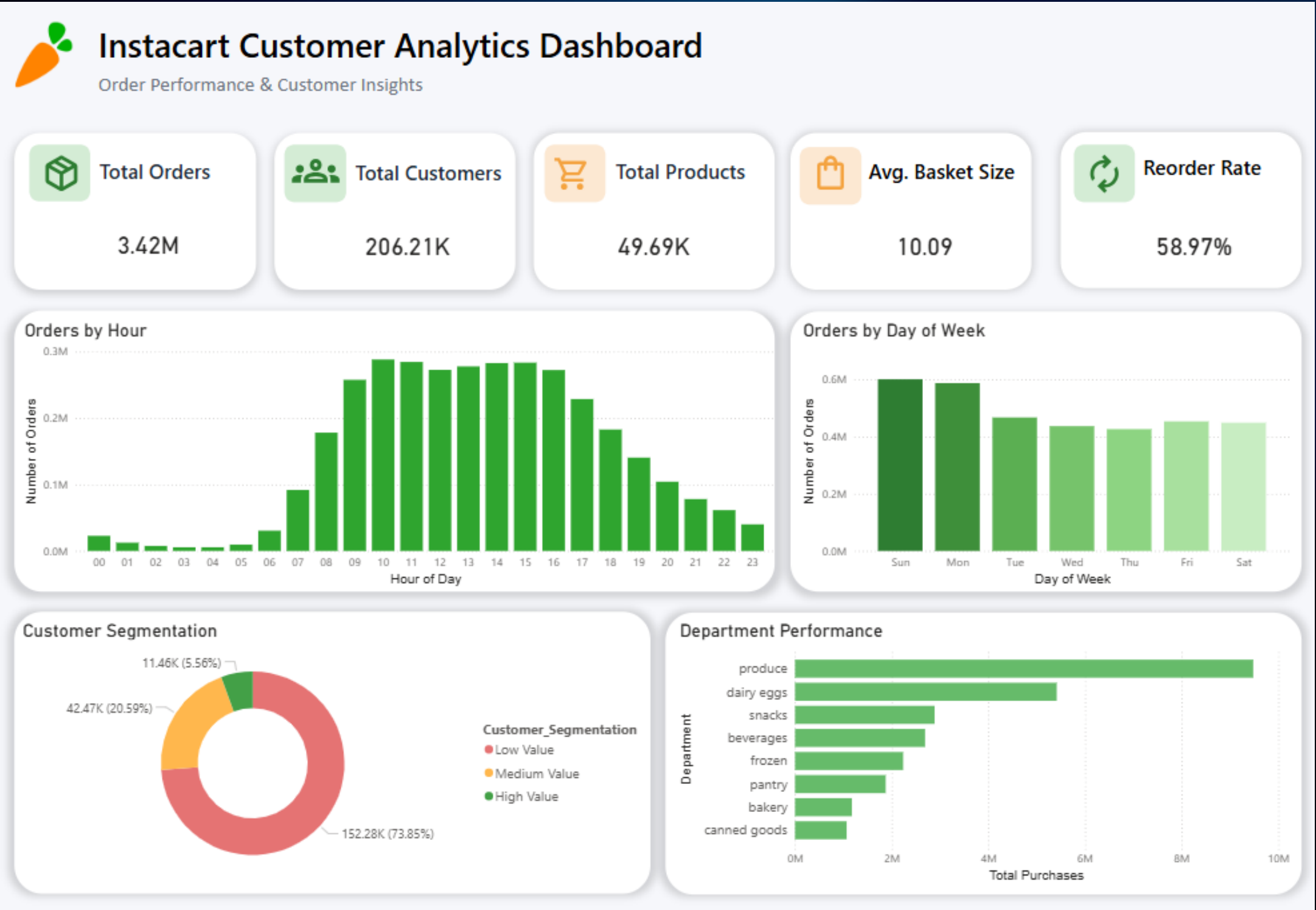
4

Customer Segmentation

How are customers grouped? Segmented by ordering behavior.

SQL transformed raw transactional data into reusable, business-ready analytical views powering the Power BI dashboards.

Executive Dashboard



Dashboard Purpose

Provides an executive overview of overall business performance at a glance.

Key KPIs

- Total Orders
- Total Customers
- Total Products
- Average Basket Size
- Overall Reorder Rate

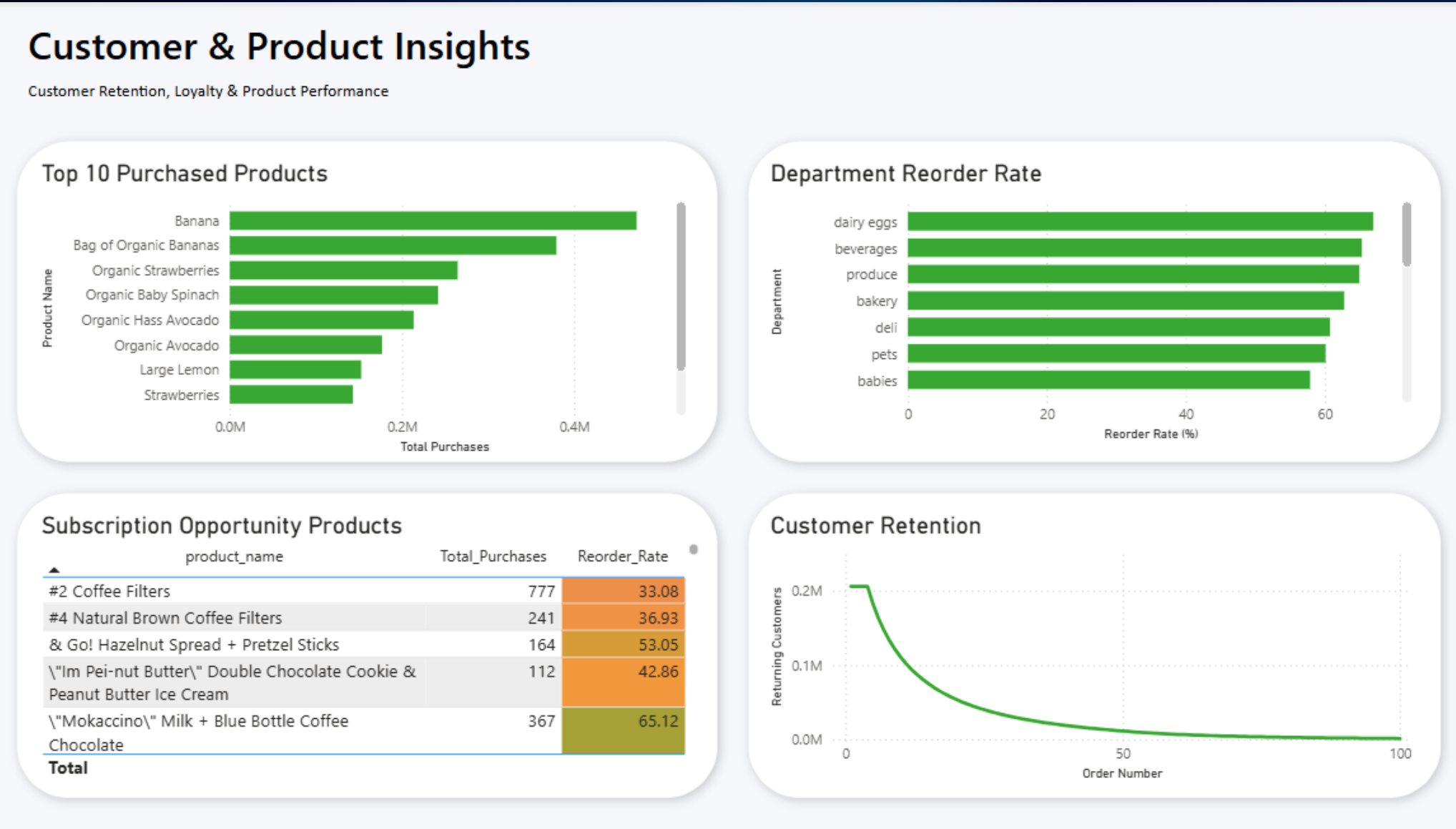
Core Visuals

- Orders by Hour
- Orders by Day of Week
- Customer Segmentation
- Department Performance



Enables quick executive decision-making with real-time KPI tracking.

Customer & Product Insights



Key Visuals

Business Value

- Improve inventory planning
- Increase customer retention
- Identify recurring revenue opportunities
- Support targeted marketing campaigns

● Top Purchased Products

Ranked by frequency and volume

● Department Reorder Rate

Loyalty indicators by category

● Subscription Opportunity Products

High-reorder candidates flagged

● Customer Retention Trends

Retention curves over time

Key Insights & Conclusion

-  **Peak Demand Timing**
Customer demand peaks during late morning and afternoon hours.
-  **High-Volume Days**
Sundays and Mondays generate the highest order volume.
-  **Top Department**
Produce is the highest-performing department by order share.
-  **Most Purchased Product**
Bananas are the single most purchased product across all orders.
-  **Subscription Candidates**
High-reorder products are ideal candidates for subscription models.
-  **Retention Opportunity**
Retention decreases over time – signaling a need for loyalty programs.

This project demonstrates a complete end-to-end analytics workflow – transforming over 3.4 million Instacart transactions into executive-level business insights using Python, DuckDB, SQL, and Power BI. It showcases practical skills in data engineering, SQL analytics, business intelligence, and dashboard design while enabling data-driven decision-making for customer behavior, product performance, and operational planning.

Thank You